Contents

The Standard Model's Structure from $(\mathbb{S}, \tau)$ — Explained Edition

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Companion exposition to the unified paper, Revision 1.2 (closed) — updated edition. This edition additionally includes, clearly marked as such, the *fourth-witness corollary* proposed for the paper's Revision 1.3: under hostile review at the time of writing, with every fact it uses already proven and twice-verified, its merge into the closed paper pending that review. Apart from that one marked block and the two definitions it requires, this document is content-identical to the closed paper in every claim, number, theorem, and open question. It changes nothing and adds nothing except *definitions*: every technical term is defined, formally and in dependency order, before it is used. It is written to be readable in principle by a person with no physics or advanced mathematics background who is willing to work — stringent, not simplified. Where the original paper states a result in one sentence, this edition states the same result in the same sentence, preceded by whatever definitions that sentence needs. Claim-status tags are used as in the whole corpus: **[FACT, proven]** marks a statement established by proof and/or verified computation; **[THM]** marks a named theorem; claims of verification refer to the two-implementation record described in Part I, §15.

Part I. The Language

§1. Vector spaces and algebras

A **real vector space** is a set of objects (called vectors) that can be added together and rescaled by real numbers, subject to the familiar rules of arithmetic (addition is commutative and associative; rescaling distributes over addition). Its **dimension** is the number of independent directions it contains: the plane has dimension 2, ordinary space dimension 3, and one writes $\mathbb{R}^n$ for the space of lists of n real numbers, which has dimension n.

An **algebra** is a vector space equipped additionally with a *multiplication*: a rule assigning to any two vectors x, y a product xy, required to be **bilinear** (it distributes over addition and rescaling in each slot separately). Nothing else is assumed. In particular, an algebra's multiplication need not be **commutative** ($xy = yx$ can fail) and need not be **associative** ($(xy)z = x(yz)$ can fail). An algebra is **unital** if it contains an element 1 with $1x = x1 = x$ for all x. A **subalgebra** is a subspace closed under the multiplication.

A **linear map** between vector spaces is a function respecting addition and rescaling. Fixing a list of independent directions (a **basis**), a linear map is recorded as a **matrix**: a rectangular array of numbers whose entry in row i, column j says how much of direction i the map produces from direction j. Composition of linear maps corresponds to **matrix multiplication**, which is associative but generally not commutative. The **rank** of a collection of vectors or matrices is the dimension of the space they span. An **eigenvector** of a linear map L is a nonzero vector v with $Lv = \lambda v$ for a number λ, its **eigenvalue**; the **spectrum** is the multiset of eigenvalues with their multiplicities; the **kernel** is the space of vectors sent to zero. The **trace** tr of a square matrix is the sum of its diagonal entries; it is independent of the chosen basis.

§2. The real division algebras and the octonions

An **inner product** $\langle x, y\rangle$ on a real vector space is a symmetric bilinear pairing with $\langle x, x\rangle > 0$ for $x \neq 0$; the associated **norm** is $|x| = \sqrt{\langle x, x\rangle}$. A **normed division algebra** is a unital algebra with an inner product satisfying $|xy| = |x||y|$: multiplication exactly respects length, and consequently a product of nonzero elements is never zero. A classical theorem (Hurwitz) states that exactly four such algebras exist over the reals: the real numbers $\mathbb{R}$ (dimension 1), the complex numbers $\mathbb{C}$ (dimension 2), the **quaternions** $\mathbb{H}$ (dimension 4, multiplication no longer commutative), and the **octonions** $\mathbb{O}$ (dimension 8, multiplication no longer associative either).

Each is obtained from the previous by the **Cayley–Dickson doubling construction**, which builds a $2n$-dimensional algebra out of pairs of elements of an n-dimensional one, with a specific twisted multiplication rule. The octonions have a standard basis $e_0 = 1, e_1, \dots, e_7$, where e_0 is the unit and

$e_1, \ldots, e_7$ each square to -1 and anticommute in pairs ($e_ie_j = -e_je_i$ for $i \neq j \geq 1$). The span of $e_1, \ldots, e_7$ is the **imaginary part** $\mathrm{Im}(\mathbb{O})$, dimension 7; the **conjugate** $\bar{x}$ of an octonion flips the sign of its imaginary part; $\langle x, y\rangle$ is the real part of $x\bar{y}$.

Doubling once more produces the sixteen-dimensional **sedenions** $\mathbb{S}$, which are *not* a division algebra: nonzero elements can multiply to zero. $\mathbb{S}$ splits as $\mathbb{O} \oplus \mathbb{O}e_8$ (pairs of octonions), and τ denotes the involution — the self-inverse linear symmetry — fixing the first copy and negating the second. The pair $(\mathbb{S}, \tau)$ is this research corpus's foundational object; the present paper uses two of its consequences: the octonions as the fixed half, and, in Part II §5, the negated half $\mathbb{O}e_8$ with the multiplication $\mathbb{S}$ itself induces on it.

§3. Symmetries: automorphisms, derivations, Lie algebras

An **automorphism** of an algebra is an invertible linear map ϕ preserving the multiplication: $\phi(xy) = \phi(x)\phi(y)$. The automorphisms of an algebra form a **group**: a set with an associative composition, an identity, and inverses. Groups of this kind, whose elements vary continuously, are **Lie groups**.

A **derivation** is the infinitesimal version of an automorphism: a linear map D obeying the product rule $D(xy) = D(x)\,y + x\,D(y)$. The derivations of an algebra form a **Lie algebra**: a vector space equipped with a **bracket** $[X, Y] = XY - YX$ (for maps, the commutator), which is antisymmetric and satisfies the **Jacobi identity** $[X, [Y, Z]] + [Y, [Z, X]] + [Z, [X, Y]] = 0$. Lie algebras are the computational skeletons of Lie groups; this paper works almost entirely at the Lie-algebra level, written in fraktur letters ($\mathfrak{g}$, $\mathfrak{su}(3)$, …). A Lie algebra is **closed** if brackets of its elements stay inside it; its **derived algebra** is the span of all brackets; its **center** is the set of elements bracketing to zero with everything.

The derivation Lie algebra of the octonions is the 14-dimensional **exceptional Lie algebra** $\mathfrak{g}_2$ — "exceptional" because it belongs to the short, sporadic list ($\mathfrak{g}_2, \mathfrak{f}_4, \mathfrak{e}_6, \mathfrak{e}_7, \mathfrak{e}_8$) of simple Lie algebras outside the infinite classical families.

Two constructions recur constantly. The **stabilizer** of an element v (inside a given Lie algebra of maps) is the subalgebra of maps sending v to zero — infinitesimally, the symmetries *fixing* v. The **centralizer** of a subalgebra $\mathfrak{h}$ is the set of elements bracketing to zero with all of $\mathfrak{h}$ — the symmetries *commuting with* $\mathfrak{h}$.

§4. Jordan algebras and $J_3(\mathbb{O})$

A **Hermitian matrix** over the octonions is a 3×3 array whose diagonal entries are real numbers and whose off-diagonal entries are octonions, with the entry at position (j, i) the conjugate of the entry at (i, j). These form a 27-dimensional real vector space: 3 diagonal reals plus 3 independent octonionic slots of 8 dimensions each.

Ordinary matrix multiplication does not keep this space closed, but the **symmetrized product** $x \circ y = \frac{1}{2}(xy+yx)$ does. Equipped with $\circ$, the space becomes the **exceptional Jordan algebra** (Albert algebra) $J_3(\mathbb{O})$: a **Jordan algebra**, meaning $\circ$ is commutative and satisfies the Jordan identity $(x^2 \circ y) \circ x = x^2 \circ (y \circ x)$, a weakened associativity. For each element x, L_x denotes the linear map "multiply by x": $L_x(y) = x \circ y$.

The **trace** of an element is the sum of its three diagonal entries; the **trace form** $\langle x, y \rangle := \mathrm{tr}(x \circ y)$ is an inner product on the 27-dimensional space and is the canonical metric used throughout this paper. J_0 denotes the 26-dimensional subspace of trace-zero elements.

The automorphism group of $J_3(\mathbb{O})$ is the exceptional Lie group F_4; its derivation Lie algebra $\mathfrak{f}_4$ has dimension 52. Because octonion-slot symmetries preserve the Jordan product, $\mathfrak{g}_2$ sits inside $\mathfrak{f}_4$ (acting in each octonionic slot simultaneously).

§5. Idempotents and the Peirce decomposition

An **idempotent** is an element A with $A \circ A = A$ — the algebraic abstraction of a projection. It is **primitive** if it cannot be written as a sum of two nonzero idempotents that multiply to zero against each other (**orthogonal** idempotents). In $J_3(\mathbb{O})$, primitive idempotents are the rank-one projections; the diagonal matrices E_{11}, E_{22}, E_{33} (a single 1 on the diagonal) are the standard examples, and a **primitive idempotent frame** is a triple of pairwise-orthogonal primitive idempotents summing to the identity.

For an idempotent A, the multiplication map L_A has eigenvalues only in $\{0, \frac{1}{2}, 1\}$; the resulting eigenspace splitting of the 27-dimensional space is the **Peirce decomposition**, written $J_0(A) \oplus J_{\frac{1}{2}}(A) \oplus J_1(A)$, and the eigenvalue attached to a state is its **Peirce weight**. For a primitive A the dimensions are 10, 16, 1.

§6. Representations

A **representation** of a Lie algebra $\mathfrak{g}$ is a vector space V together with an assignment of a linear map $\rho(X)$ on V to each $X \in \mathfrak{g}$, respecting brackets: $\rho([X, Y]) = \rho(X)\rho(Y) - \rho(Y)\rho(X)$. One says $\mathfrak{g}$ **acts** on V, and calls V a **module**. A subspace is **invariant** if the action maps it into itself; a representation with no invariant subspaces except 0 and V is **irreducible** — an unbreakable unit of symmetry action. Physicists' multiplets are exactly irreducible representations.

An **intertwiner** (equivariant map) between two representations is a linear map commuting with the action; $\mathrm{Hom}(V, W)$ denotes the space of intertwiners, and $\dim \mathrm{Hom} = 0$ means the two representations share no common piece — they are **inequivalent**. **Schur's lemma** states that the **commutant** of an irreducible action — the maps commuting with everything the algebra does — is as small as the field allows; a commutant of dimension 1 certifies irreducibility. Every representation of the compact Lie algebras used here decomposes into a direct sum of irreducibles; the **multiset** of which irreducibles appear, with multiplicities, is its complete fingerprint.

The **conjugate** $\bar{V}$ of a complex representation is the same space with the action complex-conjugated. A representation equivalent to its own conjugate is **self-conjugate**; the paper's phrase "a state plus its mirror" refers to a representation appearing together with its conjugate, which is the unavoidable format when complex representation content is carried inside a *real* vector space.

§7. Cartan subalgebras, weights, charges

A **Cartan subalgebra** (here, **torus** at the Lie-algebra level) of a compact Lie algebra is a maximal subalgebra of mutually commuting elements; its dimension is the **rank**. Commuting operators can be simultaneously diagonalized: a representation splits into joint eigenvectors, and the list of eigenvalues a joint eigenvector carries — one per Cartan generator — is its **weight vector**, physically its list of **charges**. Classifying a representation and reading off all its charges are the same computation.

Names used throughout: $\mathfrak{u}(1)$ is the one-dimensional Lie algebra (a single charge); $\mathfrak{su}(2)$ is the three-dimensional simple Lie algebra whose irreducible representations have dimensions $1, 2, 3, \ldots$ — called **singlet, doublet, triplet**, …; $\mathfrak{su}(3)$ is the eight-dimensional simple Lie algebra whose smallest faithful representations are the three-dimensional **3** (**fundamental**) and its conjugate $\bar{\mathbf{3}}$, with the eight-dimensional **adjoint 8** (a Lie algebra acting on itself by brackets is the **adjoint representation**). In $\mathfrak{su}(2)$, the Cartan generator is written T_3; the doublet's T_3-eigenvalues are $\pm\frac{1}{2}$, the triplet's $0, \pm 1$; the **raising and lowering** elements $T_\pm$ move between weight states.

The **Casimir operator** of a compact Lie algebra action is the canonical quadratic expression $\sum_{ij}(K^{-1})_{ij}\rho(X_i)\rho(X_j)$ built with the inverse of the **Killing form** $K_{ij} = \mathrm{tr}(\mathrm{ad}_{X_i}\mathrm{ad}_{X_j})$ (the natural inner product a Lie algebra carries on itself, computed from its own bracket action $\mathrm{ad}_X = [X, \cdot\,]$). The Casimir commutes with the whole action and is therefore constant on each irreducible piece — a label reading off which multiplet a state belongs to. For $\mathfrak{su}(2)$, its value on the spin-j multiplet is proportional to $j(j+1)$; the triplet-to-doublet ratio is $\frac{2}{3/4} = \frac{8}{3}$, a number the paper uses as a checkpoint.

The Killing form's **signature** — how many of its eigenvalues are positive, negative, zero — distinguishes the **real forms** of a complex Lie algebra: all-negative signature means **compact** (the associated group is closed and bounded, all its representations behave like rotations); mixed signature means a **noncompact real form**. The 78-dimensional $\mathfrak{e}_6$ has, among others, the noncompact form $\mathfrak{e}_{6(-26)}$ (signature -26, i.e. 26 positive and 52 negative directions) and the compact form (signature -78).

Branching means restricting a representation of a larger algebra to a subalgebra and re-decomposing into the subalgebra's irreducibles; the resulting multiset is the **branching multiset**. Two embeddings of the same subalgebra into the same larger algebra are **equivalent** (conjugate) if an inner symmetry of the larger algebra carries one onto the other; equivalent embeddings force identical branching multisets, so *distinct* branching multisets prove embeddings inequivalent — a fact the paper's final theorem turns into a weapon.

§8. The Standard Model, as representation data

The **Standard Model** of particle physics describes matter and three of the four known forces. For this paper only its *kinematic* layer matters, and that layer is exactly the language above: a Lie algebra of symmetries, and matter organized into its representations.

The **gauge Lie algebra** is $\mathfrak{su}(3)_{\text{color}} \oplus \mathfrak{su}(2)_{\text{weak}} \oplus \mathfrak{u}(1)_Y$. The $\mathfrak{su}(3)$ factor is the symmetry of the **strong force**; its charge is **color**, and matter is a color **triplet** (a **quark**), antitriplet, or **singlet** (a **lepton**, blind to the strong force). The $\mathfrak{su}(2)$ factor drives the **weak force** (responsible for radioactive beta decay); its Cartan charge is **weak isospin** T_3, and matter comes in weak doublets and singlets. The $\mathfrak{u}(1)$ charge is **hypercharge** Y. The observed **electric charge** is the combination $Q = T_3 + Y$ (in the normalization used here). One **generation** of matter, in the convention where all particles are listed by their left-handed components: a quark doublet ($Y = \frac{1}{6}$), a lepton doublet ($Y = -\frac{1}{2}$), and weak singlets u_R, d_R, e_R (conventional names for the right-handed up quark, down quark, and electron, listed via conjugates at $Y = -\frac{2}{3}, +\frac{1}{3}, +1$). Since the overall scale of Y is a convention, the physical content is the *ratios*; this paper's **ratio units** divide all hypercharges by the quark doublet's $\frac{1}{6}$, so the generation reads: quark doublet 1, lepton doublet -3, d_R-type ± 2, u_R-type ± 4, e_R-type ± 6.

Chirality ("handedness") is, formally, a $\mathbb{Z}_2$ grading of matter — a splitting into two halves, called left and right — with the empirically central property that the weak force treats the halves differently (**parity violation**). In the Standard Model, left-handed fermions form weak doublets while right-handed ones are weak singlets. A matter spectrum whose representation content is self-conjugate (every multiplet accompanied by its conjugate) is called **vectorlike**; one that is not is **chiral**. Nature's spectrum is chiral.

Grand Unified Theories (GUTs) embed the Standard Model algebra into a single larger simple algebra; the classic chain uses $\mathfrak{so}(10)$ (the Lie algebra of rotations of 10-dimensional space) and $\mathfrak{e}_6$, under which one generation famously organizes into $\mathfrak{so}(10)$'s 16-dimensional **spinor** representation, with $\mathfrak{e}_6$'s smallest nontrivial representation — the complex **27**, which is exactly the complexified $J_3(\mathbb{O})$ — decomposing as $\mathbf{16} \oplus \mathbf{10} \oplus \mathbf{1}$.

§9. Clifford algebras, spinors, chirality operators, triality

Given an n-dimensional real vector space with a quadratic form (an inner product, possibly with signs), its **Clifford algebra** is the algebra generated by the vectors subject only to the relation $\{v, w\} := vw + wv = \pm 2\langle v, w\rangle$ — the sign convention fixing the algebra's **signature**. $\mathrm{Cl}(0, 8)$ denotes the Clifford algebra of eight generators $\Gamma_1, \dots, \Gamma_8$ each squaring to -1 and pairwise anticommuting: $\{\Gamma_i, \Gamma_j\} = -2\delta_{ij}$. Clifford algebras are the natural home of **spinors** — the representations on which "rotations" act through the Clifford structure — and of chirality.

Products of an even number of generators span the **even subalgebra**. The **volume element** is the ordered product $\Omega = \Gamma_1 \cdots \Gamma_n$ of all generators. When n is even, Ω anticommutes with every generator; if additionally $\Omega^2 = +1$, its ± 1 eigenspaces split the spinor space into two halves that every generator swaps — Ω is then a **chirality operator**, and the two halves are the two chiralities. When n is odd, Ω is central and no such splitting exists. Evenness, not any finer detail, is the mechanism.

The even symmetries of $\mathrm{Cl}(0, 8)$ form the Lie algebra $\mathfrak{so}(8)$ (28-dimensional; infinitesimal rotations of 8-space, representable as antisymmetric 8×8 matrices). $\mathfrak{so}(8)$ has three famous 8-dimensional irreducible representations — the **vector** (ordinary rotation action) and the two chiral **half-spinors** — which are pairwise inequivalent yet completely symmetric among themselves under an outer symmetry: this phenomenon is **triality**, and proving it concretely means exhibiting matched structure constants together with $\dim \mathrm{Hom} = 0$ between all three. $\mathfrak{spin}(7) \subset \mathfrak{so}(8)$ is the rotation algebra of 7-space acting on an 8-dimensional spinor; the subalgebra of $\mathfrak{spin}(7)$ fixing one spinor is precisely $\mathfrak{g}_2$ — one of the octonions' many appearances in spin geometry.

§10. Complex and quaternionic structures

A **complex structure** on a real vector space is a linear map J with $J^2 = -\mathrm{Id}$: it lets real dimensions pair up into complex ones, with J playing the role of multiplication by i. A **quaternionic structure** is a triple of complex structures I, J, K pairwise anticommuting with $IJ = K$: it organizes the space as a module over the quaternions, written $\mathbb{H}^n$ when of quaternionic dimension n. The Lie algebra $\mathfrak{sp}(1)$ — abstractly the same as $\mathfrak{su}(2)$ — is the algebra of imaginary quaternions acting by multiplication; a representation carrying such an action *is* a quaternionic module.

Complexification means extending scalars from real to complex numbers: a real n-dimensional space becomes a complex n-dimensional one, written $V \otimes \mathbb{C}$ or here $\mathbb{C}^{27}$. The complexified space remembers its real origin through a **real structure** (conjugation): the map C fixing the original real vectors and reversing the sign of i; symmetries that were real to begin with commute with C, and a representation obtained by complexifying a real one in this way is said to be of **real type** — it is automatically self-conjugate, since C itself supplies the equivalence with the conjugate. A **polarization** of a real space carrying a complex structure J (Part I §10 below) is the projection onto J's $+i$ eigenspace inside the complexification — the choice of "half," the positive-frequency side in physics usage. Two elementary facts about polarizations are used below: conjugation always exchanges the two halves ($CPC^{-1} = 1 - P$ for the projection P), so *every* polarization breaks the real structure; and the selected half contains exactly one member of each conjugate pair, so a polarization halves the representation content pair by pair. Real symmetric operators keep real eigenvalues under complexification; real antisymmetric operators acquire purely imaginary ones; multiplying an antisymmetric operator by i makes it **Hermitian** (equal to its conjugate transpose), the complex condition guaranteeing real eigenvalues and orthogonal eigenvectors.

§11. The specific named objects of this construction

The **mediator** e is a fixed unit imaginary octonion (concretely e_4), whose left-multiplication supplies $\mathrm{Im}(\mathbb{O})$'s complex structure in this corpus's separate quantization program — the reason it was load-bearing before any of the present paper existed. The **idempotent** is

$$A = \begin{pmatrix} 0 & 0 & 0 \\ 0 & \frac{1}{2} & \frac{1}{2}e \\ 0 & \frac{1}{2}\bar{e} & \frac{1}{2} \end{pmatrix},$$

abbreviated $(0, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}e, 0, 0)$: the unique primitive idempotent the mediator determines in the chosen matrix corner. $A_0 := A - \frac{1}{3}\mathbb{1}$ is its trace-free part. E_1', E_2' denote the primitive sub-idempotents refining $\mathbb{1} - A$, so that $\{A, E_1', E_2'\}$ is a primitive idempotent frame. In the computationally convenient **aligned frame** one takes $A = E_{33}$, justified without loss of generality because F_4 acts transitively on primitive idempotents (any one can be carried to any other by an automorphism, and every claim made is conjugation-invariant).

§12–15. The verification vocabulary

Machine precision means agreement at the level of floating-point arithmetic's intrinsic resolution, around 10^{-15} in the computations here — the signature of an exact identity computed numerically, as opposed to an approximate one. **Exact (symbolic) arithmetic** means computing with fractions as fractions, no rounding at all. **Two independent implementations** means every census-level number was computed twice, by two differently-built software stacks sharing no code, no data structures, and where possible not even the same construction route, with agreement demanded on every convention-independent quantity (**invariants**: dimensions, ranks, exact fractions, ratios, Hom-dimensions, branching multisets) and every convention-dependent one reconciled by explicitly declared unit systems. **Pre-registration** means the expected outcome of a computation was written down, exactly, before the computation ran — the discipline that separates prediction from after-the-fact pattern-finding. A **spectral gap**, reported alongside every dimension claim obtained from numerical linear algebra, is the ratio between the smallest quantity treated as nonzero and the largest treated as zero; gaps of eight or more orders of magnitude accompany every such claim here. The complete checkpoint tables, error ledger, and process history live in the corpus's development notes, deliberately outside the papers.

Part II. The Paper

(From here on, this edition states exactly the content of the unified paper, Revision 1.2, section by section, in its original order and at its original claim-strength, with explanatory prose drawing only on Part I.)

1. Introduction

Dubois-Violette and Todorov showed the Standard Model gauge algebra arises inside $\mathfrak{f}_4 = \mathrm{Der}(J_3(\mathbb{O}))$ as the intersection of two natural subalgebras; Boyle recast this in idempotent-stabilizer form; Baez and Schwahn proved the general intersection theorem; Krasnov gave a $\mathrm{Spin}(9)$ characterization. This paper reports an independent arrival at that structure by a route the published program does not use — the mediator and idempotent of §11 above, objects this corpus needed for unrelated reasons — and then continues past it: to the chirality mechanism the sedenions' own split supplies, to a theorem identifying that mechanism's carrier with the weak sector's matter module, and to the exact boundary of what $J_3(\mathbb{O})$ and its complexification contain.

Two standing structural facts frame everything, both elementary and both proven in the working notes: neither $\mathbb{S}$ nor $J_3(\mathbb{O})$ embeds in the other (a dimension count blocks one direction; commutativity of the Jordan product blocks the other), and $J_3(\mathbb{O})$ is adopted from the literature, not derived from $\mathbb{O}$ — the bridge between the two constructions lives at the level of shared symmetries, $\mathfrak{g}_2 \subset \mathfrak{f}_4$, and, as §6 below proves, at the level of one shared module.

Conventions. The trace form of Part I §4 is the metric throughout; degenerate spectra are only ever interrogated with canonical (trace or Killing) metrics — a hard rule this project's error ledger repeatedly justified. Hypercharge-type quantities are quoted in the ratio units of Part I §8 unless marked raw, and every reported convention-dependent number carries its unit system.

2. Color: the mediator's $\mathfrak{su}(3)$

[FACT, proven] $\mathfrak{g}_2$ is 14-dimensional; the stabilizer of the mediator within it (Part I §3: the derivations annihilating e) is exactly 8-dimensional and acts on the six non-mediator imaginary directions as the genuine fundamental representation of $\mathfrak{su}(3)$ — irreducible by Schur's lemma, commutant of dimension 1 — with respect to the complex structure $J = L_e$ that the mediator itself supplies (Part

I §10: six real directions pairing into three complex ones). Embedded slot-wise into $\mathfrak{f}_4$ (and verified to be a genuine derivation of the Jordan product to machine precision in both implementations), this is, by construction, the published program's **color** $\mathfrak{su}(3)$: the stabilizer, inside G_2, of a fixed unit imaginary octonion. Color's triplet dimension — the "three" of three quark colors — is here literally the three complex dimensions the mediator carves out of the seven imaginary octonionic directions.

3. The second $\mathfrak{su}(3)$, $\mathfrak{e}_{6(-26)}$, and electroweak breaking

[FACT, proven] $\mathfrak{f}_4$ has dimension 52 (computed as the null space of the full derivation-condition system, a 10206×729 linear problem, in both implementations). The centralizer of color inside $\mathfrak{f}_4$ (Part I §3) is exactly 8-dimensional, closed under brackets, with negative-definite Killing form — hence a second, independent $\mathfrak{su}(3)$: the **electroweak** $\mathfrak{su}(3)$, the reservoir from which the weak force and hypercharge will be carved.

[FACT, proven] Adjoining the trace-free multiplication operators $L(J_0)$ to $\mathfrak{f}_4$ yields $\mathfrak{e}_6$: closure follows from two classical identities — a derivation bracketed with a multiplication gives the multiplication by the derived element, $[D, L_x] = L_{D(x)}$, and two multiplications bracket to a derivation, $[L_x, L_y] \in \mathrm{Der}$ — each stated and machine-confirmed. The Killing signature is exactly -26: 26 positive directions (the multiplications) against 52 negative (the derivations) — the noncompact real form $\mathfrak{e}_{6(-26)}$ with maximal compact subalgebra $\mathfrak{f}_4$ (Part I §7). The compact form $\mathfrak{f}_4 \oplus i\,L(J_0)$ has signature exactly $(0, 78, 0)$, verified from independently assembled structure constants.

[FACT, proven] A is a primitive idempotent: Peirce spectrum $\{0, \frac{1}{2}, 1\}$ at multiplicities $\{10, 16, 1\}$ (both implementations). Its stabilizer within the electroweak $\mathfrak{su}(3)$ is exactly 4-dimensional and splits as Part I §3 predicts a reductive stabilizer must: a 1-dimensional center — the hypercharge $\mathfrak{u}(1)_Y$, verified to commute with all of color to 10^{-16}, i.e. **hypercharge is color-blind**, as in nature — and a 3-dimensional derived algebra, the weak $\mathfrak{su}(2)$. (A rival candidate — the 3-dimensional centralizer of all of $\mathfrak{g}_2$ in $\mathfrak{f}_4$ — is disqualified by its own representation content: acting on the 26-dimensional trace-free space it produces exclusively triplets and quintets, never the doublets the weak force requires; recorded for completeness.)

4. Three branching theorems and the exact charges

Theorem 1 (ratio 3). Under the branching $\mathfrak{su}(3) \supset \mathfrak{su}(2) \oplus \mathfrak{u}(1)$ with hypercharge direction $Y = \mathrm{diag}(a, a, -2a)$ (the general color-blind form), the doublets inside the *adjoint* representation carry hypercharge $\pm 3a$ against the *fundamental's* doublet at a: a pure Lie-theoretic ratio of -3. This was measured first ($0.343/0.1143 = 3.00$ in the raw computation) and proven afterward. The Standard Model's own lepton-to-quark doublet hypercharge ratio, $(-\frac{1}{2})/(\frac{1}{6}) = -3$, is this number: in this construction, leptons sit in the electroweak adjoint's doublets and quarks in fundamental doublets, and the famous ratio stops being an input.

Theorem 2 (8/1/7, with exact pre-registered predictions). The 26-dimensional trace-free space decomposes under color×electroweak as $26 = (\mathbf{1}, \mathbf{8}) \oplus (\mathbf{3}, \mathbf{3}) \oplus (\bar{\mathbf{3}}, \bar{\mathbf{3}})$ — each pair being (color representation, electroweak-$\mathfrak{su}(3)$ representation), so $(\mathbf{3}, \mathbf{3})$ is a color triplet in the electroweak *fundamental*, not a weak triplet. Breaking the electroweak $\mathfrak{su}(3)$ to $\mathfrak{su}(2) \oplus \mathfrak{u}(1)$ then *forces* exactly 8 weak doublets, 1 weak triplet, and 7 weak singlets, and predicts — written in exact form before the computation ran — colored singlets at hypercharge ratio -2, one color-singlet at 0, and a state at ratio $+4$ *forbidden*. All confirmed, including the required absence.

Theorem 3 (Peirce grade is weak type). Taking $A = E_{33}$ without loss of generality (Part I §11), an element's Peirce weight counts how many of its matrix indices equal 3, while the weak $\mathfrak{su}(2)$ rotates indices $\{1, 2\}$; a state is a weak doublet exactly when it carries one index from each world — which is verbatim the Peirce-$\frac{1}{2}$ condition. The physical statement: **which states feel the weak force as doublets is the idempotent's own eigenvalue geometry.** Verified three independent ways (three constructions of the same 16-dimensional subspace, identical by rank).

[FACT, verified twice] The complete census of the 26: six colored-doublet pairs, two lepton-doublet pairs (ratio ± 3), three colored-singlet pairs (ratio ∓ 2), one sterile singlet (ratio 0), one neutral weak triplet. Every multiplicity and every ratio is a branching consequence — **zero free parameters in this census**. With $T_3 = \pm\frac{1}{2}$ on doublets (definitional, Part I §7) and the single overall calibration $Y(\text{colored doublet}) = \frac{1}{6}$, the electric charge $Q := T_3 + Y$ lands **every one of the 26 states on a Standard Model value**, $Q \in \{0, \pm\frac{1}{3}, \pm\frac{2}{3}, \pm 1\}$, with the neutral triplet's own $\{0, \pm 1\}$ matching the W-boson pattern, consistent with its gauge-adjoint reading. The conjugate doubling throughout — each entry a state plus its mirror — is the real-representation necessity of Part I §6.

5. Chirality from $\mathbb{S}$'s own odd half

[FACT, proven] The sedenions' negated half $\mathbb{O}e_8$ carries, under $\mathbb{S}$'s own multiplication, a uniform Clifford structure of signature $(0, 8)$: writing elements as pairs, $\{(0, x), (0, y)\} = -2\langle x, y\rangle \cdot 1$ holds across **all eight** directions including the real one ($(0, 1)^2 = -1$; defect exactly 0 over all 64 generator pairs, both implementations). The sedenions supply $\mathrm{Cl}(0, 8)$ — an even Clifford algebra — for free.

[FACT, proven] Seven generators alone give a *central* volume element: no chirality, structurally (Part I §9). All eight give a genuine chirality operator: $\Omega^2 = +1$, eigenvalues ± 1 at multiplicities $8/8$, every generator swapping the two halves exactly. **The mechanism is evenness, not signature.** What the direction $(0, 1)$ specifically contributes is the completion of an odd seven to an even eight — and it is the *unique* direction fixed by all of $\mathfrak{g}_2$ (every derivation kills the unit), which is why the lifted $\mathfrak{g}_2$, and color inside it, provably never mix the two chiral halves.

[FACT, proven] Each half is the 8-dimensional spinor representation of $\mathfrak{spin}(7)$ — the representation whose stabilizer of a single spinor is G_2 itself (Part I §9). The full even symmetry is $\mathfrak{so}(8)$, closure exact at dimension 28, and the two halves together with the vector representation realize complete, textbook **triality**: pairwise $\dim \mathrm{Hom} = 0$, structure constants matched across all three at 10^{-16}, verified in two unrelated constructions of the Clifford algebra (a matrix-tensor model and an octonion-multiplication model). Color's action on the two halves is *identical* — the strong force comes out **vectorlike**, parity-conserving, exactly as observed for the strong interaction in nature.

6. The bridge theorem

[FACT, proven] The stabilizer of A inside all of $\mathfrak{f}_4$ is exactly 36-dimensional — the dimension of $\mathfrak{spin}(9)$ — contains the weak $\mathfrak{u}(2)$ (one line: commuting with L_A forces annihilating A), and acts

irreducibly on the 16-dimensional Peirce-$\frac{1}{2}$ space $J_{\frac{1}{2}}$: Schur commutant exactly 1, Casimir uniform across all sixteen states at the exact value $|9/14|$ in intrinsic Killing normalization.

[FACT, proven] The nine trace-free directions of the Peirce-0 space satisfy a native Clifford relation *on* $J_{\frac{1}{2}}$: $\{L_u, L_v\}|_{J_{1/2}} = \frac{1}{4}\langle u, v\rangle$ Id, exhaustively exact over all 81 pairs — the gauge side's own spinor structure, a direct analogue of §5's. Within the octonionic slot among those nine, the analogues of $\mathbb{O}e_8$'s $(0,1)$ and of the mediator are identified by *the identical invariant on both sides*: full-$\mathfrak{g}_2$-fixedness selects the slot's real unit r (fixed exactly) against the slot's mediator m (moved) — the same criterion that distinguishes $(0,1)$ in §5. The two **chiral columns** are the sub-idempotents' Peirce-$\frac{1}{2}$ eigenspaces (each 8-dimensional, coinciding with the ninth generator's $\pm$ eigenspaces by rank); all eight slot generators are *purely* column-swapping, both diagnostics exactly zero.

[THM, the bridge] Define the dictionary φ sending the slot's octonion content to $\mathbb{O}e_8$: $\varphi(x) \leftrightarrow (0, x)$. Then: the composite J — equal to $4L_m L_r$ for raw slot units, $8L_{\hat{m}} L_{\hat{r}}$ for unit-normalized ones; the coefficient is normalization-dependent and both are stated — is a genuine complex structure on the columns, commuting with color; the color-centralizer of the gauge side's even algebra is exactly 2-dimensional (forced by Schur's lemma through the identical argument on both sides: color acts on a column as $\mathbf{1} \oplus \mathbf{1} \oplus \mathbf{3} \oplus \bar{\mathbf{3}}$) and equals span$\{J,\ \text{the pairing sum}\}$, mapping under φ to the $\mathbb{O}e_8$ side's pair $\{\Gamma_{\text{med}}\Gamma_{\text{real}},\ J_{\text{part}}\}$ **exactly**. Because the gauge side's Clifford constant is positive-definite while $\mathbb{O}e_8$'s is negative-definite, full identification over the reals is *not* the correct claim; the proven statement is: **same module, same even structure, opposite real signature.** Every load-bearing piece — the complex structure, the $\mathfrak{so}(8)$ action, the forced two-dimensional centralizer, $\mathfrak{g}_2$-compatibility — lives in the even subalgebra, which is insensitive to the signature, and matches exactly under φ; the single residue, the sign of the real quadratic form, is confined to the odd generators' squares and stated rather than absorbed.

7. Weak isospin *is* the transported chirality: $T_3 = \lambda J$ and the quaternionic module

[FACT, proven] The prerequisite identity holds exactly: the ninth generator's action on $J_{\frac{1}{2}}$ is proportional to the ordered product of the other eight ($\Gamma_9 \propto \Omega_8$, the classical mechanism by which an even Clifford algebra's volume becomes the next odd one's extra generator) — so the two columns genuinely *are* the transported image of $\mathbb{O}e_8$'s chirality splitting, and the question of this section is posed on the right object. Hypercharge preserves both columns at 10^{-16}; the raising and lowering generators $T_\pm$ — constructed canonically as the Killing-orthogonal complement of the Cartan, never by numerical search — are *purely* column-mixing, exactly.

[THM] $T_3 = \lambda J$ **identically** on $J_{\frac{1}{2}}$: the difference is 10^{-16}, with $\lambda^2 = \frac{1}{24}$ exactly ($\lambda = 1/(2\sqrt{6})$ to thirteen digits) in the aligned normalization. In words: **weak isospin's own Cartan generator is a scalar multiple of the bridge's complex structure** — $J_{\text{weak}} = J_{\text{bridge}} = 4L_m L_r$, one operator arrived at independently from the electroweak stabilizer chain and from the $\mathbb{O}e_8$ dictionary. Moreover T_3 has a closed form: $T_3 = 2\lambda[L_m, L_r]$, the idempotent frame's own inner derivation (residual 8×10^{-16}; independently anchored by the defining annihilation property that extracts the Cartan).

[THM] $J_{\frac{1}{2}}$ is a **quaternionic module**, $\mathbb{H}^4$, with the weak $\mathfrak{su}(2)$ acting as $\mathfrak{sp}(1)$ (Part I §10): $T_\pm^2 = -\mu^2$ Id, all four anticommutators $\{T_+, T_-\}$, $\{J, T_\pm\}$ vanish at the numerical floor, and the closure bracket $[T_+, T_-] = (2\mu^2/\lambda)\, T_3$ holds as an exact identity in each implementation's own normalization — the two pipelines' different μ^2 values ($\frac{1}{26}$ and $\frac{1}{24}$) both satisfy it, proving that mismatch purely conventional. The transported chirality operator is the quaternionic i; the column-mixing by $T_\pm$ is quaternion multiplication, nothing more exotic.

The mechanical fact. Weak isospin acts *across* the transported chirality split; hypercharge acts *along* it. What this means physically is deliberately held open in §11.

8. The rank-6 torus: every generator named

[FACT, proven] The operator $i\,L_{A_0}$ commutes with the entire Standard Model chain — all of color, T_3, and Y, at 10^{-16} — giving a fifth commuting charge beyond the Standard Model's rank-4 Cartan. The centralizer of that Cartan within the compact form's $i\,L(J_0)$ is exactly 2-dimensional, and the sixth, final torus direction is $i\,L_{E_1'-E_2'}$ — the multiplication by the *sub-idempotent difference*, which is simultaneously the ninth Clifford generator of §6: one object holding two jobs. The complete rank-6 torus of $\mathfrak{e}_6$: two color Cartan directions; $T_3 = 2\lambda[L_m, L_r]$; Y (the stabilizer's center); $i\,L_{A_0}$; $i\,L_{E_1'-E_2'}$. **The two directions beyond the Standard Model are the trace-free diagonal multiplications of the complete primitive idempotent frame** $\{A, E_1', E_2'\}$ — so every generator in the entire construction is a named, closed-form function of the mediator and the idempotent frame; nothing anonymous remains anywhere in the chain. A bonus identity: the census's sterile singlet *is* A_0 — the unique color-and-weak singlet of the 26 and the idempotent's own trace-free direction are one object.

9. The 27-census and the lattice point

[THM, pairing] The census space for the six charges is the full complexified $\mathbb{C}^{27}$ (Part I §10) — the new charges do not even preserve the trace-free 26. There, realized as commuting Hermitian operators (Part I §10: $i\times$antisymmetric for the $\mathfrak{f}_4$-side charges, symmetric as-is for the multiplication charges $Q_A :=$ eigenvalue of L_{A_0} and $Q_w :=$ eigenvalue of $L_{E_1'-E_2'}$), the six charges are simultaneously diagonalizable with **27 distinct weight vectors**: **twelve conjugate pairs**, on which — a theorem of the symmetric/antisymmetric split, verified for all twelve — the multiplication charges are *equal* and the $\mathfrak{f}_4$-side charges (color, T_3, Y) are *negated*; plus **three self-conjugate states**, which are **the idempotent frame itself**: A at $(Q_A, Q_w) = (\frac{2}{3}, 0)$, and E_1', E_2' at $(-\frac{1}{3}, \pm 1)$, verified by joint eigenvalue equations. Q_A's full spectrum is exactly the Peirce weights. The 27 contains its own measuring frame as three of its states, and the old split into "trace-free 26 plus a separate singlet" does not survive once the full torus acts.

[FACT, proven, exact arithmetic] Consider extended hypercharges $Y' = \alpha Y + \beta Q_A + \gamma Q_w$, all quantities in $\frac{1}{6}$-units for Y and Q_A (Q_w at its natural $\pm\frac{1}{2}$); the defining equations are stated with their numeric coefficients so the convention travels inside them. The d_R-type pair carries $(Y_r, Q_{A,r}) = (\mp 2, -2)$, so holding it at -2 while its conjugate partner reaches the u_R slot $+4$ reads $-2\alpha - 2\beta = -2$ and $+2\alpha - 2\beta = +4$, forcing $\alpha = \frac{3}{2}$, $\beta = -\frac{1}{2}$ exactly. (In the mixed convention with Q_A raw, the same physics gives $\beta = -3$; the invariant is the contribution $\beta\, Q_A(d_R) = +1$, identical in both.) Downstream, in the same units: the lepton doublet carries $(Y_r, Q_{A,r}) = (-3, 1)$, so under the quark anchor $Y'(L) = \frac{3}{2}(-3) - \frac{1}{2}(1) = -5$. Without the u_R demand, the system holding quarks, leptons, and d_R at Standard Model values has the **unique** solution $(\alpha, \beta, \gamma) = (1, 0, 0)$ — the original hypercharge, where u_R was absent. At the u_R-forcing point, γ splits by anchor: the quark sector demands $\gamma = 0$ (a chirality-blind extended hypercharge), the lepton sector $\gamma = \pm 4$ (chirality-sensitive per column); jointly the system is over-determined. u_R**'s price**: whichever sector is not held fixed is displaced by exactly **two ratio units** — leptons to -5 as computed above, or quark columns to $\frac{3}{2}(1) - \frac{1}{2}(1) \pm 4 \cdot \frac{1}{2} = 1 \pm 2 = \{3, -1\}$ — symmetric and exact. And e_R's slot is **structurally empty**: the census contains *no color-singlet weak-singlet pair at all* — the twelfth pair is the weak triplet's charged states.

10. The inequivalence theorem

[THM] The embedding of the Standard Model algebra into E_6 selected by this construction is **inequivalent** to the $SO(10)$-GUT embedding (Part I §7–8). Witnesses, all computed: (i) this construction's 27 carries a weak-*triplet* matter state; the GUT 27 has none; (ii) it carries six colored weak-doublet pairs — three colors times two chiralities — where the GUT 27 carries three; (iii) the sector trade-off of §9, which the GUT embedding does not exhibit. The theorem is then internalized: the centralizer of $i\, L_{A_0}$ in compact $\mathfrak{e}_6$ is exactly $46 = 36 + 9 + 1 = \mathfrak{so}(10) \oplus \mathfrak{u}(1)$; all 46 generators respect the Peirce splitting at 10^{-16}, so **the** $\mathfrak{so}(10)$ **spinor sixteen and the Peirce-**$\frac{1}{2}$ **sixteen are the same subspace** in this construction — resolving a long-standing internal caution about "two different sixteens" (the GUT route's sixteen belongs to a *different, conjugate* $\mathfrak{so}(10)$ and is a different subspace); and the corpus's weak $\mathfrak{su}(2)$, sitting inside this $\mathfrak{so}(10)$, branches the blocks as $\mathbf{10} \to \mathbf{3} \oplus 7 \cdot \mathbf{1}$ and $\mathbf{16} \to 8 \cdot \mathbf{2}$ (Killing-normalized Casimir; the triplet-to-doublet Casimir ratio landing at $8/3$ exactly, per Part I §7) against the GUT class's hand-derived $\mathbf{10} \to (\mathbf{2}, \mathbf{2}) \oplus 6 \cdot \mathbf{1}$ and $\mathbf{16} \to 4 \cdot \mathbf{2} \oplus 8 \cdot \mathbf{1}$: **distinct branching multisets in both representations, hence distinct conjugacy classes of** $\mathfrak{su}(2)$ **inside one and the same** $\mathfrak{so}(10)$ — the sole external input being that two-line textbook branching (Slansky).

Consequently: the absence of e_R, the price of u_R, and the presence of the triplet are not failures of this construction to reach the Standard Model — they are exact features of the embedding its own objects canonically select, and the precise location at which the published program's complexified and bi-Cayley extensions become necessary rather than optional.

[Corollary — the fourth witness; Revision 1.3 candidate, under review at the time of this edition; all facts used are proven and twice-verified.] The census's Standard-Model-side weight content — the (color, T_3, Y) values over all 27 states — is exactly symmetric under negation: every state's mirror is present, verified directly from the census data, and disclosed in kind by §4's own "each entry is a state plus its mirror." This is structurally forced, not incidental: the embedding factors through F_4, which is precisely the stabilizer of the 27's real structure (Part I §10), so its action on $\mathbb{C}^{27}$ is the complexification of a real representation — of real type, hence **vectorlike** under the Standard Model algebra (Part I §8): it admits no chiral generation and no parity violation, and the missing e_R is one symptom of this deeper shape. The contrapositive is the witness: any *chiral* branching of the same 27 — the GUT class's, or the published program's bi-Cayley construction — provably does **not** factor through this F_4; the chiral-versus-vectorlike contrast is the inequivalence theorem's fourth and most physical witness. Two boundary statements then hold at full strength. First, no representation-level device escapes the vectorlike verdict from inside: a polarization (Part I §10) necessarily breaks the real structure and halves each conjugate pair, so half-*selection* is the one place chirality could enter — and which half gets selected is state-level data, outside this paper's scope by its own §11. Second, the charge inventory binds every such half: **a slot absent from the**

census is absent from all of its halves, so e_R's absence constrains any state-selected spectrum this 27 can ever yield, polarized or not.

11. Discussion and scope

Relation to the published program. The destination — the Standard Model algebra from $J_3(\mathbb{O})$ — is Dubois-Violette–Todorov's, in Boyle's idempotent-stabilizer form, generalized by Baez–Schwahn. What is this corpus's own: (i) a derivation, rather than an assumption, of *why octonions* — a separately-published enumeration theorem proving the relevant non-associative structure forced; (ii) the mediator as the single shared origin of both color and complexification; (iii) the bridge and $T_3 = \lambda J$ theorems, connecting the gauge structure to a chirality mechanism the program does not have; (iv) the inequivalence theorem and the boundary results of §§9–10, which sharpen the program's own stated limits into exact statements.

The fork, held open by design. The transported chirality grading turns out to be the weak $\mathfrak{su}(2)$'s own Cartan axis — an object with no spacetime (Lorentz) pedigree. *Reading A (transmuted)*: the weak sector does not couple to a pre-existing chirality; it *contains* one, as an axis of its own quaternionic action. *Reading B*: the physical analogue of the spacetime chirality operator γ^5 is a different object, living, if anywhere in this corpus, at the fermionic state level where spacetime structure actually enters. The γ-ambiguity of §9 is prior to this question, not a resolution of it. Nothing in this paper decides the fork; the two readings are recorded at equal rank, with Reading B's evidence currently heavier.

Open, named. Canonical selection: whether the construction's own objects select a γ-anchor (hence a chirality-sensitivity) rather than leaving it external. The electroweak mixing angle: a structural reason for the Weinberg combination beyond $Q = T_3 + Y$ landing correctly. The fermionic program — quantum states on the $\mathrm{Cl}(0,8)$ module, a state-dependent chirality-odd conjugation, the quantization of this census — is the named frontier, deliberately not begun here.

What was not attempted. Gauge dynamics, a Lagrangian, the Higgs mechanism, particle masses, generations, or any claim beyond the group-, representation-, and module-theoretic content above. This is a mathematics paper about physics-shaped structure, in the same genre as the published program it extends.

Verification statement. Two independent implementations (different octonion arithmetic, different representations of $J_3(\mathbb{O})$, different extraction routes; the Clifford side built by matrix-tensor and octonion-multiplication models respectively), agreeing on every invariant checkpoint — dimensions, ranks, exact fractions, ratios, Hom-dimensions, branching multisets — with all convention-dependent scalars reconciled by stated unit systems, and the lattice results in exact rational arithmetic. Pre-registered expectation lists preceded every computation stage on both sides; the complete checkpoint tables, the deviations ledger, and the process history are in the development notes.

References

- Companion papers of this corpus: the enumeration paper (*The Non-Associativity Selector*); $J_3(\mathbb{O})$*: Construction, the Peirce Split, and Verified Frame Relativity via* F_4; *From* $J_3(\mathbb{O})$ *to* $E_7(-25)$; the Foundation and Keystone papers for $(\mathbb{S}, \tau)$, the mediator, and the quantization

chain.
- Dubois-Violette, M., “Exceptional quantum geometry and particle physics,” *Nucl. Phys. B* 912 (2016) 426–449.
- Todorov, I. & Dubois-Violette, M., “Deducing the symmetry of the standard model from the automorphism and structure groups of the exceptional Jordan algebra,” *Int. J. Mod. Phys. A* 33 (2018) 1850118, arXiv:1806.09450.
- Dubois-Violette, M. & Todorov, I., “Exceptional quantum geometry and particle physics II,” *Nucl. Phys. B* 938 (2019) 751–761, arXiv:1808.08110.
- Boyle, L., “The standard model, the exceptional Jordan algebra, and triality,” *J. Math. Phys.* 67 (2026) 071701.
- Baez, J. & Schwahn, P., “The Standard Model Gauge Group from the Exceptional Jordan Algebra,” arXiv:2606.15235.
- Krasnov, K., “$SO(9)$ characterisation of the standard model gauge group,” arXiv:1912.11282.
- Slansky, R., “Group theory for unified model building,” *Phys. Rep.* 79 (1981) 1–128.
- Baez, J. C., “The Octonions,” *Bull. Amer. Math. Soc.* 39 (2002) 145–205.
- Chevalley, C. & Schafer, R. D., “The Exceptional Simple Lie Algebras F_4 and E_6,” *Proc. Nat. Acad. Sci. USA* 36 (1950) 137–141.

Created in collaboration with Claude (Anthropic). All conclusions and any errors remain the author’s responsibility.